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EDUCATION

Expected Jan 2027 **SABANCI UNIVERSITY - *B.S. in Computer Science, Faculty of Engineering***
GPA: 3.84/4.00 **Minors:** Business Analytics and Mind & Behavior
Honors/Awards: Seven terms High Honor Certificate (GPA higher than 3.5), %100 Scholarship (Outstanding Achievement Scholarship)

SKILLS

Programming: C++ (Advanced), HTML, CSS, JavaScript, SQL, NodeJS, React, Dart, Verilog, Python, Laravel
Language: Turkish (Native), English (Advanced), German (Beginner)

WORK EXPERIENCE

Nov – Present **Hepsiburada Istanbul, Turkey – *Data Science Intern***
- Worked on LLM-based chatbot systems, focusing on improving response diversity while maintaining factual consistency and relevance across multi-turn conversations.
- Developed techniques to reduce hallucinations and topic drift in long-context chatbot interactions, improving conversational stability and coherence.
- Conducted experiments on query autocomplete ranking, addressing position bias to improve fairness and relevance in ranked suggestions.

Jun – Aug 2025 **Albasoft Information Technologies Ankara, Turkey – *Full-Stack Developer Intern***
- Developing a Hospital Information Management System. Using Laravel with MVC architecture.
- Enhanced user interface and responsiveness using Bootstrap and implemented interactive features with jQuery. Gaining experience in full-stack web development in a real-world setting.

Feb – Jun 2025 **Sabanci University Istanbul, Turkey – *Data Structures Learning Assistant***
- Assisted students with understanding and solving data structures problems, providing clear explanations and support, and hold office hours.

Feb - Jun 2024 **Sabanci University Istanbul, Turkey – *Calculus-II Learning Assistant***
- Solved Calculus-II problems for freshman students and provided clear explanations to enhance their understanding of problem-solving techniques.

Jan - Feb 2024 **TUBITAK BILGEM Istanbul, Turkey - *Semester break internship***
- Gained experience in working with technologies and systems used in radio signal applications.
- Solved data structures and algorithm tasks using C++. Gained hands-on experience in hardware design with Verilog. Applied problem-solving skills in both software and hardware domains.

PROJECTS

May – Jun 2026 **SuSchedule-r, LLM-Based Course-Planning Agent,** *A grounded, tool-using course-planning agent, built with a 3-person team.*

- Built a GPT-4o ReAct agent for Sabancı University course planning, where the LLM handles language while deterministic tools own factual constraints (prerequisites, requirements, conflicts).
- Designed a two-stage RAG system (BAAI/bge-base-en-v1.5 + FAISS) over 817 courses and a 22-tool toolbox grouped by responsibility (eligibility, requirements, catalog search, timetabling).
- Evaluated across four axes with strong results — 0.977 Hit@5 on retrieval and 91.7% end-to-end grounding — alongside perfect conflict detection and near-exact degree-requirement matching.

Apr – May 2026 **Eliminating Position Bias in Query Autocomplete Ranking,** *A search-ranking fairness project at Hepsiburada.*

- Analyzed click-through-rate behavior by ranking position on Hepsiburada's query autocomplete, quantifying a strong position bias (CTR dropping from 51% at rank 1 to 1% at rank 8) that causes models to learn position popularity rather than true relevance.
- Implemented Inverse Propensity Weighting (IPW) for stable exposure-bias correction, then Doubly Robust (DR) learning, and identified an over-correction failure mode where DR excessively suppressed high-CTR keyword-type results.
- Designed a score-blending approach combining baseline and DR scores to preserve keyword relevance while promoting underexposed item types, producing a fairer, more interpretable ranking.

Apr – May 2026 **HepsiBuddy, AI-Powered Shopping Assistant,** *A concept developed for a Hepsiburada ideathon competition.*

- Designed an AI-powered shopping assistant concept for a company-wide Hepsiburada ideathon, building an interactive HTML prototype with a custom mascot to demonstrate the conversational shopping flow.
- Produced a 14-15 slide presentation with revenue modeling (~34M TL/month) and competitive positioning against Amazon Rufus and Allegro's assistant, plus scripts and Q&A prep for the pitch.

Mar – Apr 2026 **Content Viral, YouTube-to-TikTok Automation Pipeline,** *A full-stack video repurposing and clip-generation platform.*

- Built an end-to-end full-stack pipeline (Next.js 14, FastAPI, Celery/Redis, SQLite) that discovers trending YouTube videos and auto-converts them into vertical TikTok/Reels-ready clips with AI-generated captions and hashtags.
- Used yt-dlp/ffmpeg for video processing and the OpenAI API (Whisper + GPT) for AI-driven clip segmentation, plus a movie-split mode, a Creative-Commons-filtered discovery page, and a clip review/edit interface.

Feb – Mar 2026 **SubsAI, Subscription Tracking Platform,** *An AI-powered personal finance management project.*

- Built a full-stack subscription tracking application using Next.js 14, FastAPI, and PostgreSQL for persistent data storage.
- Integrated the OpenAI API to automatically detect, categorize, and summarize recurring subscriptions from bank statement data, reducing manual entry.
- Designed a RESTful API architecture connecting the FastAPI backend to the Next.js frontend, handling authentication, subscription CRUD operations, and billing-cycle logic.
- Implemented features for tracking renewal dates, spending trends, and cost breakdowns, giving users actionable insights into recurring expenses.

Jan – Feb 2026 **MovieRAG, Retrieval-Augmented Generation Pipeline,** *A movie question-answering system.*

- Built a Retrieval-Augmented Generation (RAG) pipeline over the TMDB 5000 movie dataset to enable accurate, context-grounded movie question-answering.
- Implemented a two-stage retrieval architecture using a BAAI bi-encoder for initial candidate retrieval and a cross-encoder reranker to refine relevance ranking.
- Integrated Qwen2.5-3B-Instruct as the generation model to produce natural-language answers grounded in retrieved movie context.
- Conducted systematic failure case analysis (FC1-FC5) and ablation studies, identifying and debugging retrieval and generation errors to improve pipeline robustness.

Nov – Dec 2025 **Song2Vec, Music Embedding & Recommendation System,** *A Word2Vec-based playlist embedding project.*

- Trained skip-gram Word2Vec embeddings on large-scale playlist data, built a preprocessing pipeline and analyzed vocabulary/frequency distributions.
- Ran a hyperparameter grid search selecting the best configuration via Hit Rate@k/NDCG@k, and validated it through song-algebra experiments and t-SNE visualization.
- Built a recommendation interface with cold-start handling and a structured failure analysis identifying four error modes: artist over-representation, cold-start failures, popularity bias, and genre drift.

Oct – Nov 2025 **CNN-based Smile Detection from Celebrity Images,** *Machine Learning project.*

- Developed a Convolutional Neural Network (CNN) to classify smiling vs. non-smiling facial expressions from celebrity face images.
- Performed data preprocessing and normalization on image datasets, including resizing, grayscale/RGB handling, and train–validation splitting.
- Designed and trained CNN architectures with convolutional, pooling, and fully connected layers, optimizing performance through hyperparameter tuning.
- Evaluated model performance using accuracy and loss metrics and analyzed misclassifications.

June – Sep 2025 **Hospital Information Management System,** *A full-stack role-based Hospital Management platform.*

- Developed core modules including appointment scheduling, patient records, doctor approval system, and medication tracking using Laravel, MySQL, Bootstrap, and jQuery.
- Applied MVC architecture, middleware-based access control, and secure user authentication.
- Developed admin approval flow, dynamic records, and scheduling features for role-specific users.

Feb – May 2025 **Digital E-commerce Website,** *A full-stack Online Clothing Store developed as a team project.*

- Built a fully functional e-commerce website for clothing sales using React for the frontend and Node.js with Express for the backend.
- Implemented MongoDB for product, user, cart, and order data storage, with Mongoose for schema management. Enabled user authentication and role-based access control (e.g., admin, customer).

Feb – May 2025 **Sabancı Sepeti,** *Designed and built a cross-platform Food Ordering and Delivery mobile app.*

- Built with Flutter and Dart, used Firebase Authentication for secure user login and registration. Connected to Firebase Firestore for real-time data handling.
- Managed state efficiently with the Provider package. Designed UI/UX using Figma, ensuring a clean and modern look. Implemented form validation, navigation, and responsive layouts.

CERTIFICATIONS

Sep 2024 **A1.1 German Certificate, University of Wien**
Jul 2024 **Young Executive Academy, Global Executive Academy**
Oct 2023 **Python Basics, University of Michigan**
Sep 2023 **Essential Management Skills, International Business Management Institute**
Jan 2021 **The Duke of Edinburgh Bronze Award, The Duke of Edinburgh's International Award Foundation**

VOLUNTARILY WORK

Sep 2018 - Present **TEMA FOUNDATION, Associate**

- Contributed many tree planting activities.

Feb - Jun 2023 **Sabanci University Civic Involvement Projects, Volunteer**

- Visited the nursing home once a week for 2 hours to spend time with the residents in there.

May 2023 **Box Observer at Turkey's Presidential Election**

- Been constantly around the ballot box to follow and supervise all transactions, to check whether they recorded correctly or not and been ready to object if necessary.